

Bài tập cuối 11

5.5.

Sách Pinchover - Rubinstein

Ticung 124 - 127

Bài 5.5.

$$u_t - k u_{xx} = 0, \quad 0 < x < L, \quad t > 0.$$

$$u_x(0, t) = u_x(L, t) = 0, \quad t \geq 0.$$

$$u(x, 0) = f(x), \quad 0 \leq x \leq L.$$

a) Ta xác định cơ sở của không gian nghiệm của bài toán gồm các phân tử có dạng

$$u(x, t) = X(x) \cdot T(t)$$

Từ ptt. $u_t = k u_{xx}$

$$\Rightarrow X(x) \cdot T'(t) = k X''(x) \cdot T(t)$$

$$\Rightarrow \frac{T'(t)}{k T(t)} = \frac{X''(x)}{X(x)} = \text{const}$$

Kết hợp với đk $u_x(0, t) = u_x(L, t)$, Ta có bài toán Sturm-Liouville

$$\begin{cases} X''(x) - \text{const} X(x) = 0 \\ X'(0) = X'(L) = 0 \end{cases}$$

$$\text{Xét } \int_0^L X(x) [X''(x) - \text{const} X(x)] dx = 0$$

$$\Leftrightarrow \int_0^L X(x) d(X'(x)) - \int_0^L \text{const} X^2(x) dx = 0$$

$$\Leftrightarrow X(x) X'(x) \Big|_0^L - \int_0^L [X'(x)]^2 dx - \int_0^L \text{const} X^2(x) dx = 0$$

$$\text{Do } X'(0) = X'(L) = 0$$

$$\Rightarrow \int_0^L \{ [X'(x)]^2 + \text{const} [X(x)]^2 \} dx = 0$$

$$\text{+). Nếu } \text{const} > 0 \rightarrow X'(x) = 0$$

$$\rightarrow X(x) = 0 \rightarrow u \text{ tầm thường}$$

+) Nếu $\text{const} = 0 \rightarrow$ Các giá trị riêng $\lambda_0 = 0$
Hàm riêng $X_0(x) = 1$

+) Nếu $\text{const} = -k^2 < 0$

$$\rightarrow X''(x) + k^2 X(x) = 0$$

$$\rightarrow X(x) = a \cos kx + b \sin kx$$

$$X'(x) = k(b \cos kx - a \sin kx)$$

$$X'(0) = b k \cos kx \quad b k = 0 \Rightarrow b = 0$$

$$\Rightarrow X(x) = a \cos kx$$

$$X'(L) = -a k \sin kL = 0$$

$$\rightarrow \sin kL = 0$$

$$\rightarrow kL = n\pi$$

$$\rightarrow k = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots$$

$\rightarrow$ Các giá trị riêng $\text{const} = -\left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \dots$

Hàm riêng tương ứng $X_n(x) = \cos \frac{n\pi x}{L}$

$$\text{Cđ: } \frac{T'(t)}{kT(t)} = \text{const} \Rightarrow \frac{T'(t)}{T(t)} = -k \frac{n^2 \pi^2}{L^2}$$

$$\rightarrow \frac{T(t)}{T_0} = \exp\left(-k \frac{n^2 \pi^2}{L^2} t\right)$$

$$\rightarrow \left\{ \text{cđ là hàm phân tử} \right\} \exp\left(-k \frac{n^2 \pi^2}{L^2} t\right) \cos \frac{n\pi x}{L}$$

$$\rightarrow U(x, t) = \left(\sum_{n=1}^{\infty} a_n \exp\left(-k \frac{n^2 \pi^2}{L^2} t\right) \cos \frac{n\pi x}{L} \right) + \frac{a_0}{2}$$

Trong đó

$$U(x, 0) = \left(\sum_{n=1}^{\infty} a_n \cos \frac{n\pi x}{L} \right) + \frac{a_0}{2} = f(x) \rightarrow a_n = \frac{2}{L} \int_0^L f(x) \cos \frac{n\pi x}{L} dx, \quad n = 1, 2, 3, \dots$$

$$b) \quad U_t = 12U_{xx}, \quad 0 < x < \pi, \quad t > 0$$

$$U_x(0, t) = U_x(\pi, t) = 0 \quad (t \geq 0)$$

$$U(x, 0) = 1 + 8\sin^3 x \quad 0 \leq x \leq \pi$$

Theo câu b

$$U(x, t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{\frac{-2\sqrt{3}n^2\pi^2 t}{\pi^2}} \cos \frac{n\pi x}{\pi}$$

$$= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n e^{-2\sqrt{3}n^2 t} \cos nx$$

Tính để: $a_n = \frac{2}{\pi} \int_0^{\pi} (1 + 8\sin^3 x) \cos nx \, dx$

$$= \frac{2}{\pi} \int_0^{\pi} \cos nx \, dx + \frac{2}{\pi} \int_0^{\pi} 8\sin^3 x \cos nx \, dx$$

$$= \frac{2}{\pi} \left. \frac{8\sin nx}{n} \right|_0^{\pi} + \frac{2}{\pi} \int_0^{\pi} \frac{3\sin x - 4\sin 3x}{4} \cos nx \, dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} (3\sin x \cos nx - 8\sin 3x \cos nx) \, dx$$

$$= \frac{1}{2\pi} \int_0^{\pi} \frac{3}{2} (\sin(n+1)x + \sin(1-n)x) - \frac{1}{2} (\sin(3+n)x + \sin(3-n)x) \, dx$$

$$= \frac{1}{2\pi} \left[\frac{3}{2(n+1)} \cos(n+1)x + \frac{3}{2(1-n)} \cos(1-n)x \right] - \frac{1}{2} \left[\frac{\cos(3+n)x}{n+3} + \frac{\cos(3-n)x}{n-3} \right] \Big|_0^{\pi}$$

$$= \frac{1}{2\pi} \left\{ \frac{3}{2} \left[\frac{\cos(1-n)\pi}{n-1} - \frac{\cos(n+1)\pi}{n+1} \right] - \frac{1}{2} \left[\frac{\cos(3-n)\pi}{n-3} - \frac{\cos(3+n)\pi}{n+3} \right] \right\} \Big|_0^{\pi}$$

$$= \frac{1}{2\pi} \left\{ \frac{3}{2} \left[\frac{\cos(1-n)\pi}{n-1} - \frac{\cos(n+1)\pi}{n+1} \right] - \frac{1}{2} \left[\frac{\cos(3-n)\pi}{n-3} - \frac{\cos(3+n)\pi}{n+3} \right] - 3 \left(\frac{1}{n^2-1} - \frac{1}{n^2-9} \right) \right\}$$

Nếu $n = 2k+1 \Rightarrow a_{2k+1} = \frac{1}{2\pi} \left\{ \frac{3}{2} \left[\frac{\cos 2k\pi}{2k} - \frac{\cos(2k+2)\pi}{2k+2} \right] - \frac{1}{2} \left[\frac{\cos(2-2k)\pi}{2k-2} - \frac{\cos(2k+4)\pi}{2k+4} \right] \right.$

$k = 0, 1, 2, \dots$

$$\left. - 3 \left(\frac{1}{(2k+1)^2-1} - \frac{1}{(2k+1)^2-9} \right) \right\} = 0$$

Nếu $n = 2k$, $k = 1, 2, \dots$

$$\begin{aligned} \rightarrow q_{n=2k} &= \frac{1}{2\pi} \left[\frac{3}{2} \left(\frac{1}{n+1} - \frac{1}{n-1} \right) - \frac{1}{2} \left(\frac{1}{n+3} - \frac{1}{n-3} \right) - 3 \left(\frac{1}{n^2-1} - \frac{1}{n^2-9} \right) \right] \\ &= \frac{1}{2\pi} \left[\frac{6}{n^2-9} - \frac{6}{n^2-1} \right] = \frac{3}{\pi} \left(\frac{1}{n^2-9} - \frac{1}{n^2-1} \right) \end{aligned}$$

$$q_0 = \frac{1}{L} \int_0^L f(x) dx = \frac{1}{\pi} \int_0^\pi (1 + \sin^3 x) dx$$

$$= \frac{1}{\pi} \int_0^\pi \left(1 + \frac{3}{4} \sin x - \frac{1}{4} \sin 3x \right) dx = \frac{1}{\pi} \left(x - \frac{3}{4} \cos x + \frac{1}{12} \cos 3x \right) \Big|_0^\pi$$

$$= \frac{1}{\pi} \left(\pi + \frac{3}{4} - \frac{1}{12} - \pi + \frac{3}{4} - \frac{1}{12} \right) = 1 + \frac{4}{3\pi}$$

$$\rightarrow u(x,t) = \frac{q_0}{2} + \sum_{n=1}^{\infty} q_n e^{-2\sqrt{3}n^2 t} \cos nx$$

$$= \begin{cases} 1 + \frac{4}{3\pi} & \text{nếu } n = 2k+1 \quad (k = 0, 1, 2, \dots) \\ 1 + \frac{4}{3\pi} + \sum_{k=1}^{\infty} \frac{3}{\pi} \left(\frac{1}{4k^2-9} - \frac{1}{4k^2-1} \right) e^{-2\sqrt{3} \cdot 4k^2 t} \cos 2kx, & (n = 2k) \\ & k = 1, 2, \dots \end{cases}$$

b) Do $e^{-2\sqrt{3}n^2 t} \rightarrow 0$ khi $n \rightarrow \infty$

$$\Rightarrow \lim_{t \rightarrow \infty} u(x,t) = \frac{q_0}{2} = \frac{1}{L} \int_0^L f(x) dx$$

5.6.

$$u_t = k u_{xx}, \quad 0 < x < 2\pi, \quad t > 0$$

$$u(0,t) = u(2\pi,t), \quad u_x(0,t) = u_x(2\pi,t), \quad t > 0$$

$$u(x,0) = f(x), \quad 0 \leq x \leq 2\pi$$

a) Ta xác định cơ sở của $\mathcal{C}$ gần nó gồm các ptri dạng $u(x,t) = X(x)T(t)$

$$\text{Từ PT } u_t = k u_{xx} \Rightarrow \frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)} = \text{const}$$

Kh' hợp với đk (*) Ta có bài toán Sturm-Liouville

$$\int X'' - \text{const} X = 0$$

$$X(0) = X(2\pi), \quad X'(0) = X'(2\pi)$$

+) Nếu $\text{const} = k^2 > 0$

$$\rightarrow x'' - k^2 x = 0$$

$$\rightarrow x(x) = a e^{kx} + b e^{-kx}$$

$$x'(x) = k(a e^{kx} - b e^{-kx})$$

$$\begin{cases} x(0) = x(2\pi) \Rightarrow a + b = a e^{k2\pi} + b e^{-k2\pi} \\ x'(0) = x'(2\pi) \Rightarrow k(a - b) = k(a e^{k2\pi} - b e^{-k2\pi}) \end{cases}$$

$$\rightarrow a = b = 0 \rightarrow x(x) = 0$$

+) Nếu $\text{const} = 0$

$$\rightarrow x'' = 0$$

$$\rightarrow x = ax + b$$

$$x(0) = x(L) \Rightarrow b = aL + b \rightarrow a = 0$$

$$x'(0) = x'(L) \rightarrow b \in \mathbb{R}$$

$$\rightarrow \begin{cases} a = 0 \\ b \in \mathbb{R} \end{cases}$$

$$\rightarrow \begin{cases} \text{Giá trị riêng } \lambda_0 = 0 \\ \text{Hàm riêng } x_0(x) = 1 \end{cases}$$

+) Nếu $\text{const} = -k^2 < 0$

$$\rightarrow x'' + k^2 x = 0$$

$$\rightarrow x = a \cos kx + b \sin kx$$

$$x' = k(-b \cos kx + a \sin kx)$$

$$\begin{cases} x(0) = x(2\pi) \Rightarrow a = a \cos k2\pi + b \sin k2\pi = a \\ x'(0) = x'(2\pi) \Rightarrow bk = b k \end{cases}$$

$$\begin{cases} x(0) = x(2\pi) \Rightarrow a = a \cos k2\pi + b \sin k2\pi \\ x'(0) = x'(2\pi) \Rightarrow bk = k(b \cos k2\pi - a \sin k2\pi) \end{cases}$$

$$\rightarrow \begin{cases} a(1 - \cos k2\pi) - b \sin k2\pi = 0 \\ a \sin k2\pi + b(1 - \cos k2\pi) = 0 \end{cases}$$

Hệ trên có h^o c nghiệm t^hg $\Leftrightarrow$

$$\begin{vmatrix} 1 - \cos k2\pi & -\sin k2\pi \\ \sin k2\pi & 1 - \cos k2\pi \end{vmatrix} = 2 - 2\cos k2\pi = 0$$

$$\rightarrow \cos k2\pi = 1$$

$$\rightarrow k2\pi = n2\pi$$

$$\rightarrow k = n$$

$$(n = 1, 2, \dots)$$

$$\rightarrow \text{Các giá trị riêng } \text{const} = -n^2 \quad (n = 1, 2, \dots)$$

$$\text{Hàm riêng } \phi_n(x) = \cos nx, \psi_n(x) = \sin nx$$

$$\text{Ta có } \frac{T'(t)}{kT(t)} = \text{const} = -n^2$$

$$\rightarrow \frac{T'(t)}{T(t)} = -kn^2$$

$$T(t) = e^{-kn^2 t}$$

$$\rightarrow T_n(t) = a_n e^{-kn^2 t}$$

$\rightarrow$ (c) vế trái $\leq$ giá trị riêng gần 2 phần tử

$$\{ e^{-kn^2 t} (\cos nx + \sin nx) \},$$

$$\{ e^{-kn^2 t} \cos nx; e^{-kn^2 t} \sin nx \}$$

$$\rightarrow u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) e^{-kn^2 t}$$

$$\text{Ta có: } \begin{cases} a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x) dx \\ a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx \\ b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx \end{cases}$$

$$(b) \lim_{t \rightarrow \infty} u(x,t) = \frac{a_0}{2} \quad \text{Do } e^{-kn^2 t} \rightarrow 0 \text{ khi } t \rightarrow \infty$$

Bài 3.8

$$U_t - U_{xx} = e^{-t} \sin 3x \quad 0 < x < \pi, t > 0$$

$$u(0, t) = u(\pi, t) = 0 \quad t \geq 0$$

$$u(x, 0) = f(x) \quad 0 \leq x \leq \pi$$

(a) Chuỗi nghiệm của bài toán có dạng

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin nx$$

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) \sin nx = f(x)$$

$$\rightarrow T_n(0) = \frac{2}{\pi} \int_0^{\pi} \sin nx f(x) dx$$

$$u_t = \sum_{n=1}^{\infty} T'_n(t) \sin nx$$

$$u_{xx} = \sum_{n=1}^{\infty} -T_n(t) n^2 \sin nx$$

$$\rightarrow u_t - u_{xx} = \sum_{n=1}^{\infty} [T'_n(t) + n^2 T_n(t)] \sin nx = e^{-t} \sin 3x$$

$$\rightarrow \begin{cases} T'_n(t) + n^2 T_n(t) = 0 & \text{Khi } n \neq 3 \\ T'_3(t) + 9T_3(t) = e^{-t} \end{cases}$$

$$\rightarrow \left[e^{9t} T_3(t) \right]' = e^{9t} e^{-t}$$

$$\rightarrow T'_n(t) + n^2 T_n(t) = 0 \quad \text{Khi } n \neq 3$$

$$\rightarrow \left[e^{n^2 t} T_n(t) \right]' = 0$$

$$\rightarrow T_n(t) = e^{-n^2 t} T_n(0)$$

$$\rightarrow T_3(t) = T_3(0) e^{-9t}$$

$$(b) f(x) = x \sin x$$

$$\Rightarrow u(x,0) = \sum_{n=1}^{\infty} T_n(0) \sin nx = x \sin x$$

$$\Rightarrow T_n(0) = \frac{2}{\pi} \int_0^{\pi} x \sin nx \sin nx dx$$

$$\Rightarrow T_n(0) = 0 \quad \forall n \neq 1$$

$$n=1 \Rightarrow T_1(0) = \frac{2}{\pi} \int_0^{\pi} x \sin^2 x dx = \frac{1}{\pi} \int_0^{\pi} x(1 - \cos 2x) dx$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{\pi^2}{2} \right] = \frac{\pi}{2}$$

Ta có hệ:

$$\begin{cases} T_n'(t) + n^2 T_n(t) = 0 & \text{với } n \neq 3 \\ T_3'(t) + 9 T_3(t) = e^{-t} & \text{với } n = 3 \end{cases}$$

$$T_n'(t) + n^2 T_n(t) = 0$$

$$\Rightarrow T_n(t) = T_n(0) e^{-n^2 t} \Rightarrow T_n(t) = 0 \quad \text{với } n \neq \{1, 3\}$$

$$\Rightarrow T_1(t) = T_1(0) e^{-t} = \frac{\pi}{2} e^{-t}$$

Với $n=3$: $T_3'(t) + 9 T_3(t) = e^{-t}$

Ta có nghiệm $T_3(t) = a e^{-9t} + T_3^+(t)$

Với $T_3^+(t)$ là nghiệm của PT

$$T_3^{+'}(t) + 9 T_3^+(t) = e^{-t}$$

$$T_3^{+'}(t) = c e^{-t} \Rightarrow T_3^{+'}(t) = -c e^{-t}$$

$$\Rightarrow T_3^{+'}(t) + 9 T_3^+(t) = c e^{-t} = e^{-t} \Rightarrow c = \frac{1}{8}$$

$$\Rightarrow T_3^+(t) = \frac{e^{-t}}{8}$$

$$\Rightarrow T_3(t) = a e^{-9t} + \frac{e^{-t}}{8}$$

$$T_3(0) = a + \frac{1}{8} = 0 \Rightarrow a = -\frac{1}{8}$$

$$\Rightarrow T_3(t) = \frac{1}{8} (e^{-t} - e^{-9t})$$

$$T_1(t) = \frac{\pi}{2} e^{-t}$$

$$\Rightarrow u(x,t) = \frac{\pi}{2} e^{-t} \sin x + \frac{1}{8} (e^{-t} - e^{-9t}) \sin 3x$$

(d) Find u_t

$$u_t = -\frac{\pi}{2} e^{-t} \sin x + \frac{1}{8} (9e^{-9t} - e^{-t}) \sin 3x$$

$$u_x = \frac{\pi}{2} e^{-t} \cos x + \frac{3}{8} (e^{-t} - 9e^{-9t}) \cos 3x$$

$$u_{xx} = -\frac{\pi}{2} e^{-t} \sin x - \frac{9}{8} (e^{-t} - e^{-9t}) \sin 3x$$

$$\Rightarrow u_t - u_{xx} = e^{-t} \sin 3x$$

5.10

$$u_t = u_{xx} + \alpha u \quad 0 < x < 1, t > 0$$

$$u(0, t) = u(1, t) = 0 \quad t \geq 0$$

$$u(x, 0) = f(x) \quad 0 \leq x \leq 1, f \in C([0, 1])$$

$$a) \quad \alpha = -1 \quad \left\{ \begin{array}{l} u_t - u_{xx} = -u \quad 0 < x < 1, t > 0 \\ u(0, t) = u(1, t) = 0 \quad t \geq 0 \\ f(x) = x \quad 0 \leq x \leq 1, f \in C([0, 1]) \end{array} \right.$$

Chuỗi nghiệm của bài toán có dạng

$$u(x, t) = \sum_{n=1}^{\infty} T_n(t) \sin n\pi x$$

$$u(x, 0) = \sum_{n=1}^{\infty} T_n(0) \sin n\pi x = x$$

$$\Rightarrow T_n(t) = \frac{1}{2} \int_0^1 \sin n\pi x \cdot x \, dx$$

$$= \frac{-2}{n\pi} \int_0^1 x \, d(\cos n\pi x)$$

$$= \frac{-2}{n\pi} \left(x \cos n\pi x \Big|_0^1 - \int_0^1 \cos n\pi x \, dx \right)$$

$$= \frac{-2}{n\pi} \left(\cos n\pi - \frac{\sin n\pi}{n\pi} \Big|_0^1 \right)$$

$$= \frac{-2}{n\pi} \cos n\pi$$

$$u_t = \sum_{n=1}^{\infty} T'_n(t) \sin n\pi x$$

$$u_{xx} = \sum_{n=1}^{\infty} -(n\pi)^2 T_n(t) \sin n\pi x$$

$$\Rightarrow u_{xt} - u_{xx} = \sum_{n=1}^{\infty} [T'_n(t) + (n\pi)^2 T_n(t)] \sin n\pi x = -u(x, t)$$

$$\Rightarrow \sum_{n=1}^{\infty} [\bar{T}'_n(t) + (n\pi)^2 \bar{T}_n(t)] \sin n\pi x = - \sum_{n=1}^{\infty} \bar{T}_n(t) \sin n\pi x$$

$$\Rightarrow \bar{T}'_n(t) + (n\pi)^2 \bar{T}_n(t) = -\bar{T}_n(t)$$

$$\Rightarrow \bar{T}'_n(t) + [(n\pi)^2 + 1] \bar{T}_n(t) = 0$$

$$\Rightarrow \left[e^{[(n\pi)^2 + 1]t} \bar{T}_n(t) \right]' = 0$$

$$\Rightarrow e^{[(n\pi)^2 + 1]t} \bar{T}_n(t) = a_n$$

$$\Rightarrow \bar{T}_n(t) = a_n e^{[-(n\pi)^2 - 1]t}$$

$$\Rightarrow a_n = \bar{T}_n(0) = \frac{-2 \cos n\pi}{n\pi}$$

$$\Rightarrow \bar{T}_n(t) = \frac{-2 \cos n\pi}{n\pi} e^{-(n\pi)^2 t}$$

$$\Rightarrow U(x,t) = \sum_{n=1}^{\infty} e^{-(n\pi)^2 t} \frac{-2 \cos n\pi \sin n\pi x}{n\pi}$$

(b) Nếu $\alpha \leq 0$. Thay -1 bởi α trong CT nghiệm

$$\Rightarrow U(x,t) = \sum_{n=1}^{\infty} e^{-[(n\pi)^2 - \alpha]t} \frac{-2 \cos n\pi \sin n\pi x}{n\pi}$$

Do $\alpha \leq 0 \Rightarrow (n\pi)^2 - \alpha > 0$ (Do $n > 1$)

$$\Rightarrow e^{-[(n\pi)^2 - \alpha]t} \rightarrow 0 \text{ khi } t \rightarrow \infty$$

$$\Rightarrow U(x,t) \rightarrow 0 \text{ khi } t \rightarrow \infty$$

Sách Thầy Hợp - Bài 75

$$U_x = a^2 U_{xx}$$

$$U(x,0) = f(x) = \begin{cases} x & \text{ khi } 0 < x < L/2 \\ L-x & \text{ khi } L/2 \leq x < L \end{cases}$$

$$U(0,t) = 0 \quad \text{khi } t \geq 0$$

$$U(L,t) = 0 \quad \text{khi } t \geq 0$$

(Phương nghiệm của bài toán có dạng

$$U(x,t) = \sum_{n=1}^{\infty} T_n(t) \sin \frac{n\pi x}{L}$$

$$U(x,0) = \sum_{n=1}^{\infty} T_n(0) \sin \frac{n\pi x}{L} = f(x)$$

$$\Rightarrow T_n(0) = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

$$= \frac{2}{L} \int_0^{\frac{L}{2}} x \sin \frac{n\pi x}{L} dx + \frac{2}{L} \int_{\frac{L}{2}}^L (L-x) \sin \frac{n\pi x}{L} dx$$

$$+ \Rightarrow \bar{I}_1 = \frac{2}{L} \int_0^{\frac{L}{2}} x \sin \frac{n\pi x}{L} dx = -\frac{2}{L} \frac{L}{n\pi} \int_0^{\frac{L}{2}} x d\left(\cos \frac{n\pi x}{L}\right)$$

$$= -\frac{2}{n\pi} \left(x \cos \frac{n\pi x}{L} \Big|_0^{\frac{L}{2}} - \int_0^{\frac{L}{2}} \cos \frac{n\pi x}{L} dx \right)$$

$$= -\frac{2}{n\pi} \left(\frac{L}{2} \cos \frac{n\pi}{2} - \frac{L}{n\pi} \sin \frac{n\pi x}{L} \Big|_0^{\frac{L}{2}} \right)$$

$$= -\frac{2}{n\pi} \left(\frac{L}{2} \cos \frac{n\pi}{2} - \frac{L}{n\pi} \sin \frac{n\pi}{2} \right)$$

$$+ \int_0^L \frac{2}{L} (L-x) \sin \frac{n\pi x}{L} dx$$

$$= \frac{2}{L} - \frac{L}{n\pi} \int_{\frac{L}{2}}^L (L-x) d\left(\cos \frac{n\pi x}{L}\right)$$

$$= \frac{2-L}{L n\pi} \left((L-x) \cos \frac{n\pi x}{L} \Big|_{\frac{L}{2}}^L + \int_{\frac{L}{2}}^L \cos \frac{n\pi x}{L} dx \right)$$

$$= \frac{-2}{n\pi} \left(-\frac{L}{2} \cos \frac{n\pi}{2} + \frac{L}{n\pi} \sin \frac{n\pi x}{L} \Big|_{\frac{L}{2}}^L \right)$$

$$= -\frac{2}{n\pi} \left(-\frac{L}{2} \cos \frac{n\pi}{2} + \frac{L}{n\pi} \sin \frac{n\pi}{2} \right)$$

$$\Rightarrow T_n(t) = T_1 + T_2 = \frac{4L}{(n\pi)^2} \sin \frac{n\pi}{2}$$

$$U_t = \sum_{n=1}^{\infty} T_n'(t) \sin \frac{n\pi x}{L}$$

$$U_{xx} = \sum_{n=1}^{\infty} -\left(\frac{n\pi}{L}\right)^2 T_n(t) \sin \frac{n\pi x}{L}$$

$$\Rightarrow U_t - a^2 U_{xx} = \sum_{n=1}^{\infty} \left[T_n'(t) + \left(\frac{n\pi a}{L}\right)^2 T_n(t) \right] \sin \frac{n\pi x}{L} = 0$$

$$\Rightarrow T_n'(t) + \left(\frac{n\pi a}{L}\right)^2 T_n(t) = 0 \quad \forall n$$

$$\Rightarrow \left[e^{\left(\frac{n\pi a}{L}\right)^2 t} T_n(t) \right]' = 0$$

$$\Rightarrow T_n(t) e^{\left(\frac{n\pi a}{L}\right)^2 t} = a_n$$

$$\Rightarrow T_n(t) = e^{-\left(\frac{n\pi a}{L}\right)^2 t} a_n$$

$$\rightarrow a_n = T_n(0) = \frac{4L}{(n\pi)^2} \sin \frac{n\pi}{2}$$

$$\rightarrow T_n(t) = \frac{4L}{(n\pi)^2} \sin \frac{n\pi}{2} e^{-\left(\frac{n\pi a}{L}\right)^2 t}$$

$$\rightarrow u(x,t) = \sum_{n=1}^{\infty} \frac{4L}{(n\pi)^2} \sin \frac{n\pi}{2} e^{-\left(\frac{n\pi a}{L}\right)^2 t} \sin \frac{n\pi x}{L}$$